

Alžběta Prášilová

Computational Biology | Bioinformatics | Biomedical Data Analysis

Bioinformatician with an M.Sc. in Bioinformatics and research experience spanning computational oncology, biological image analysis, genomics, and quantitative analysis of experimental and clinical data. Currently working in the Petrenko Lab at BIOCEV on glioblastoma research, combining microscopy data analysis with mechanistic 3D modelling. In parallel, I contribute to research projects in plant genomics, microbiology, computational immunology, and clinical research, including several long-term collaborations. Interested in research-support roles in computational biology, bioinformatics, or biomedical data analysis.

+420 608 072 152 | alzbeta.prasilova@seznam.cz | github.com/prasilal | LinkedIn: Alžběta Prášilová | Relocating to Lausanne

CORE SKILLS

- Programming & scientific computing: Python, R, Julia, Bash, Git; HPC/cluster workflows, Linux, markdown
- Data & image analysis: statistical analysis, data preprocessing and QC, scientific visualisation, time-series analysis, microscopy image segmentation and quantitative image analysis; project experience with machine learning
- Bioinformatics: GWAS & post-GWAS workflows, genotype–phenotype associations, microbiome analysis; currently gaining experience in single-cell and spatial omics analysis
- Computational modelling: agent-based modelling, Cellular Potts model, reaction–diffusion systems, model calibration and comparison with experimental data

RESEARCH EXPERIENCE

Research Assistant — Computational Biology | Petrenko Lab, BIOCEV / Institute of Experimental Medicine, CAS

Jan 2026 – Present · Current position

- Develop computational workflows for 3D glioblastoma experiments, including microscopy image analysis, QC, growth, drug response, invasion and spatial analysis of fluorescently labelled stromal cells.
- Continue developing a 3D hybrid glioblastoma model combining agent-based cell mechanics, reaction–diffusion fields, and intracellular signalling and metabolism.
- Recently joined a research project at GeneCore, working on single-cell and spatial omics analysis (snail single-cell datasets and STOmics data from mouse heart tissue), expanding my experience with these data types and their computational workflows.

Builds on M.Sc. thesis work; 2 manuscripts in preparation.

Selected Research Collaborations

Computational Plant Genomics — Wheat GWAS | Czech Agrifood Research Center

Apr 2026 – Sep 2026 · Manuscript in preparation

Processed and integrated DArT genotype and phenotype data for winter hardiness and frost tolerance in wheat; performed GWAS and post-GWAS locus and candidate-gene prioritisation. Prepared statistical analyses, figures and summary tables for the ongoing manuscript.

Computational Immunology — Immune Dynamics & Multiscale Models | Mgr. Tomáš Bílý, MFF UK, Prague

2020 – present · Multi-year collaboration · VPH 2024 poster; VPH 2026 flash talk & poster

Co-developed spatial models of T-cell search for dendritic cells in lymph nodes, comparing random-walk and chemotactic search mechanisms and testing how different tissue conditions affect search behaviour. The collaboration later expanded into the multiscale glioblastoma modelling described above, including intracellular ODE models, surrogate modelling, and treatment-response simulations.

Quantitative Biology & Data Analysis — Photosynthesis and Microbial Physiology | Institute of Microbiology, CAS

Nov 2024 – present · 1 paper published (2026), 1 manuscript under review

- *Cyanobacterial nitrogen fixation*: performed statistical analyses for a comparative study of how nitrogen-fixing versus nitrogen-replete conditions affect photosynthetic state transitions in two unicellular cyanobacteria.
- *Iron limitation in marine phytoplankton:* analysed combined physiological and proteomic responses across eight phytoplankton species using PCA and sensitivity/robustness analyses.
- *Fluorescence spectroscopy & machine learning:* explored whether fluorescence emission spectra could predict nutrient concentrations or nutrient-limitation states; processed spectral and environmental data and tested regression, classification, dimensionality-reduction and feature-selection approaches.

Clinical & Microbiome Data Analysis — Otorhinolaryngology | 3rd Faculty of Medicine, Charles University, Prague
Feb 2025 – present · 1 paper published (2025), 1 manuscript under review

Analysed clinical and microbiome data for studies of the healthy sinonasal microbiome and chronic rhinosinusitis. Performed statistical and microbiome analyses, prepared publication figures and tables, and contributed to manuscript revisions and responses to reviewers.

Clinical Data Analysis — Neuromuscular Research | Motol University Hospital, Prague
Dec 2025 – May 2026 · 1 paper published (2026)

Performed statistical analysis of clinical and autonomic-function data in patients with Charcot–Marie–Tooth disease and related hereditary neuropathies. Analysed heart-rate variability and cardiovascular autonomic test outcomes across genotype groups and prepared outputs for manuscript publication.

Genomic Data Analyst | Mgr. Jan Janouškovec, PhD — Centre Algatech, Třeboň
Jul 2021 – Sep 2021

- Assisted in research on genomic data interpretation; developed Python scripts to automate analysis and support the bioinformatics workflow.

PUBLICATIONS & MANUSCRIPTS

Published:

- "Characteristics of a Healthy Sinonasal Microbiome" (2025)
- "Diazotrophy Affects the State Transitions in Unicellular Nitrogen-Fixing Cyanobacteria" (2026)
- "Cardiovascular autonomic neuropathy in Charcot-Marie-Tooth disease: a genotype-stratified study of PMP22 duplication and GJB1 mutation carriers" (2026)

Under review:

- "Sinonasal microbiome restructuring in Chronic rhinosinusitis with nasal polyps without corticosteroid and antibiotics exposure"
- "Species-specific responses to iron limitation in eukaryotic phytoplankton across diverse ecological origins"

In preparation (working titles)**:**

- "Genome-Wide Association Mapping and Genomic Selection for Improving Winter Hardiness and Frost Tolerance in Wheat"
- "Effects of Buparlisib on growth, invasion, and mesenchymal stromal cell spatial organization in 3D glioblastoma spheroids"
- "Multiscale computational modelling of glioblastoma spheroids integrating tumor–stroma interactions, intracellular signalling and metabolism, and drug response"

CONFERENCE PRESENTATIONS

- "Toward multiscale lymph node model: T cell search strategy study." VPH 2024 — [Poster]
- "A computational model for tumor-stromal interactions in 3D culture systems." BaCell3D 2026 — [Poster]
- "A multiscale model of glioblastoma-mesenchymal stromal cells tumor spheroids." VPH 2026 — Flash talk&[Poster]

EDUCATION

M.Sc. in Bioinformatics | Charles University, Prague
2022 – 2025

- Coursework: Machine Learning, Genomics, Structural Bioinformatics, Mathematical Modeling
- Erasmus exchange: Università di Torino, Turin (2023)
- Thesis: Data-driven computational model of 3D glioblastoma spheroid growth: the effect of mesenchymal stem cell integration

B.Sc. in Bioinformatics | Charles University, Prague
2019 – 2022

- Thesis: Tools for microbial community analysis using high-throughput amplicon sequencing data

ADDITIONAL EXPERIENCE, LANGUAGES & INTERESTS

- Singer, Petr Špaček & Band (2021–2025) — performed across Czech cities on annual tours to audiences of 1,000+
- Czech — native | English — C1 | German — A2 | Italian — A1
- **Interests**: singing, climbing, playing musical instruments, running, hiking